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Sensemaking-Informed Allocation of Organizational Learning Capabilities: A Mathematical Decision-Support Model

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Abstract: Organizational learning depends on how knowledge is acquired, shared, and integrated across organizational units, yet the allocation of learning capabilities often overlooks differences in how employees interpret information and develop shared understanding. This study develops a mathematical decision-support model for allocating organizational learning capability (OLC) dimensions across organizational units by incorporating sensemaking into resource allocation decisions. Drawing on organizational learning and sensemaking theories, a binary optimization model was formulated to maximize the weighted contribution of selected learning capability items under unit-level budget limits, an organization-wide budget constraint, and minimum learning capability requirements. Sensemaking was incorporated through a unit-specific effectiveness coefficient and a minimum capacity constraint representing collective interpretation and shared understanding. A structured scoring procedure was specified to estimate the effectiveness coefficients, and the model was solved using LINGO (a commercial optimization software package) for a hypothetical organization comprising five units. Sensitivity analyses were conducted by varying sensemaking coefficients, minimum capacity requirements, and the total budget. The computational results produced different allocations of learning capability dimensions across the five units. Changes in sensemaking coefficients altered selected allocations, particularly in Unit 2, while tighter minimum sensemaking requirements affected Unit 4. Budget reductions also changed the selected dimensions, whereas budget increases left the allocation unchanged under the examined conditions. These findings indicate that incorporating sensemaking into learning capability allocation can account for differences in organizational contexts and identify how resource constraints shape knowledge management priorities. The model provides a structured basis for comparing unit-specific learning investments and examining alternative allocation decisions, although its effectiveness in organizational practice requires empirical validation.

Keywords: Organizational learning capability (OLC); Sensemaking; Knowledge management; Resource allocation; Binary optimization; Decision support

1 Introduction

Organizational learning is central to how enterprises acquire, distribute, and apply knowledge in response to changing operational and competitive conditions. It involves adjustments to organizational practices and structures that support recovery from disruptions and sustained improvements in performance [1]. Jerez-Gomez et al. [2] describe organizational learning as a set of capabilities associated with knowledge creation, acquisition, transfer, and integration, through which organizations modify established patterns of behavior. These processes also involve knowledge generation, information dissemination, and organizational memory, allowing experience and information to be retained and used in subsequent decisions [3]. As digital technologies alter the ways in which knowledge is produced, communicated, and stored, organizations face new demands in coordinating learning activities across departments and functional boundaries [4]. The issue is therefore not simply whether an organization possesses learning capabilities, but how those capabilities can be developed and distributed in accordance with the needs of its constituent units.

Organizational learning capability (OLC) provides a basis for examining these differences. It encompasses the managerial practices, organizational resources, and collective skills through which learning is supported and translated into organizational action [5]. Ulrich et al. [6] associate learning capability with the ability of managers and organizational members to generate, communicate, and integrate ideas across internal boundaries. Subsequent research has connected OLC with innovation, organizational adaptation, and the development of capabilities needed to respond to changing business conditions [7, 8]. The breadth of this literature is also reflected in bibliometric and content analyses that document the development of OLC research across disciplines and organizational settings [9]. Although these studies establish the relevance of learning capability to organizational performance, they provide a less direct basis for deciding how limited resources should be assigned to particular learning activities within an organization.

The measurement of OLC has received considerable attention in both industrial and service settings. Jerez-Gomez et al. [2] and Hsu and Fang [10] examined OLC through constructs associated with knowledge processes and organizational performance. In a study of hospitals, Aghdasi and Khakzar Bafruei [11] reported differences among learning capability dimensions, with knowledge transfer and integration receiving the highest mean assessment, followed by systems perspective, openness and experimentation, and managerial commitment. Related investigations have identified variation in learning capability across managerial groups, industries, and organizational environments. More recently, organizational learning has been examined in relation to remote working arrangements [12], the adoption of artificial intelligence in knowledge-related activities [13], and digitally mediated learning processes [4]. These studies provide useful evidence on the measurement and organizational relevance of learning capabilities. Nevertheless, identifying the relative importance of individual dimensions does not, by itself, establish how they should be selected and implemented across organizational units when financial resources and learning requirements differ.

A further consideration concerns the way organizational members interpret learning initiatives and the information associated with them. The present study draws on sensemaking theory to examine this issue. Building on the work of Weick [14], Weick et al. [15], Maitlis and Christianson [16], and Sandberg and Tsoukas [17], sensemaking refers to the processes through which individuals and groups interpret ambiguous circumstances, develop plausible explanations, and establish shared understandings that guide action. In organizational learning, these processes influence how new information is understood, how experience is discussed, and whether knowledge can be incorporated into collective practices. Two units may therefore respond differently to the same learning initiative, even when similar resources are available. Differences in communication, shared mental models, collaborative practices, and organizational identification may affect the extent to which a particular learning capability can be put into practice. Although sensemaking provides a theoretical explanation for such differences, its role in the quantitative allocation of learning capabilities across organizational units remains insufficiently specified.

The allocation of OLC dimensions can consequently be treated as a resource allocation problem in which organizational units have different learning needs, implementation costs, and capacities for collective interpretation. Decisions must account for the expected contribution of each learning activity while respecting financial and organizational constraints [5, 18]. Research on data analytics and knowledge management has shown the importance of connecting information processing and knowledge sharing with managerial decision-making [19]. Mathematical approaches have also been used to represent organizational learning processes, including models based on self-modeling networks [20] and production dynamics [21]. Such approaches demonstrate that organizational learning can be examined through formal computational representations. The problem addressed here is more specific: determining which learning capability items should be implemented in each organizational unit when their expected effects depend partly on local sensemaking conditions. This requires a model that considers learning capability contributions, implementation costs, minimum learning requirements, and differences in sensemaking capacity within a common decision structure.

This allocation problem has direct implications for knowledge management. Learning capability dimensions influence different stages of knowledge acquisition, communication, integration, and retention. For example, dialogue and participative decision-making provide opportunities for organizational members to exchange interpretations, while knowledge transfer and integration concern the incorporation of individual and collective experience into established practices. Their relative importance may differ between units responsible for operational activities, coordination, or strategic decisions. Where resources are limited, assigning the same learning priorities to every unit may overlook these differences. Conversely, concentrating resources on selected activities without considering local communication and interpretation capabilities may restrict their practical use. A decision-support model can help knowledge management teams examine these trade-offs by identifying feasible combinations of learning capability items under specified budgets and minimum capability requirements. Such a model does not replace managerial judgment; rather, it provides a consistent basis for comparing allocation alternatives and examining how changes in organizational conditions may affect the resulting decisions.

Against this background, the present study develops a mathematical decision-support model for allocating OLC

dimensions across organizational units while accounting for differences in sensemaking conditions. The study makes two related contributions. First, it formulates the allocation of learning capability items as a binary optimization problem that considers their weighted contributions, unit-level and organization-wide budget limits, and minimum learning capability requirements. Second, it incorporates sensemaking into the formulation through an effectiveness coefficient that adjusts the contribution of learning capability items according to unit-specific conditions and a minimum capacity constraint representing the ability to support collective interpretation and shared understanding. A structured scoring procedure is specified for estimating the sensemaking coefficients, and computational experiments with a hypothetical five-unit organization are used to examine the resulting allocations and their sensitivity to selected parameters. The proposed model offers a means of connecting organizational learning and sensemaking concepts with resource allocation decisions in knowledge management. Its contribution is primarily methodological: it provides a formal representation of these relationships and a basis for examining alternative allocation decisions. The practical effectiveness of the model remains to be established through applications involving observed organizational data.

2 Organizational Learning Capability and Sensemaking

2.1 Organizational Learning Capability: Foundations and Dimensions

OLC concerns the conditions that allow organizations to learn from experience and retain what has been learned across different activities, units, and periods. Ashkenas et al. [22] associated this capability with the ability to carry knowledge beyond the circumstances in which it was initially developed. Without such capacity, organizations may continue refining established practices while finding it difficult to undertake more substantial changes. OLC therefore extends beyond individual learning to include organizational structures, managerial practices, and working relationships that support the acquisition, exchange, and application of knowledge [23–25]. More recent studies by Sanhokwe et al. [7] and Chin et al. [26] have examined the measurement and changing organizational relevance of learning capabilities, drawing attention to the need to consider how these capabilities operate under different organizational conditions.

Jerez-Gomez et al. [2] identified three considerations underlying OLC. The first concerns knowledge as an organizational resource, including its acquisition, dissemination, application, and integration into established activities. The second concerns collective identity and the recognition that organizational learning depends on cooperation among individuals and units working toward shared objectives. The third concerns the sustained commitment of time and resources required to establish learning practices that are not readily reproduced by competing organizations. Taken together, these considerations suggest that learning capability depends not only on the availability of knowledge but also on the organizational arrangements through which knowledge is interpreted, retained, and used. They also provide a basis for examining why the implementation of learning capabilities may require different priorities across organizational units.

OLC is generally treated as a multidimensional construct, although the number and definition of its dimensions vary across measurement approaches. Goh and Richards developed a 21-item instrument covering clarity of purpose and mission, leadership commitment and empowerment, experimentation and rewards, knowledge transfer, and teamwork and group problem solving [24]. Hult and Ferrell [23] proposed a 23-item measure organized around team orientation, systems orientation, learning orientation, and memory orientation. Jerez-Gomez et al. [2] identified four dimensions: managerial commitment, systems perspective, openness and experimentation, and knowledge transfer and integration. Despite differences in classification, these approaches share an interest in the organizational conditions that allow knowledge to move beyond individual experience and become part of collective practice.

Chiva et al. [27] approached OLC through five dimensions: experimentation, risk taking, interaction with the external environment, dialogue, and participative decision-making. These dimensions describe organizational practices through which employees encounter new information, exchange interpretations, and participate in learning-related decisions. Subsequent research has examined learning capabilities in relation to digital transformation [8], sustainability-oriented organizational practices [7], and technologies associated with changing production and management environments [28]. These developments indicate that the relative importance of learning capability dimensions may vary according to organizational circumstances. For the present study, the different measurement approaches provide a set of candidate dimensions for allocation decisions rather than a single, universally applicable ranking of learning priorities. The distinction is important because a dimension considered relevant to organizational learning as a whole may not require the same level of investment in every organizational unit.

2.2 Sensemaking as a Foundation for Organizational Learning

Organizational learning involves more than the acquisition and circulation of information. Employees must also interpret new knowledge, relate it to existing experience, and establish sufficient common understanding for coordinated action. Sensemaking addresses these interpretive processes, particularly when organizational members encounter unfamiliar, uncertain, or contradictory circumstances [16]. It is therefore relevant to understanding why similar learning practices may be received and implemented differently across organizational units. Recent studies

have examined sensemaking in organizational adaptation, workplace learning, and the use of language in constructing shared interpretations [29–31].

Weick [14] described seven properties of sensemaking that are relevant to organizational learning:

- Identity grounding. Individuals interpret events in relation to their professional roles, responsibilities, and perceptions of organizational membership. Consequently, employees working in different units may assign different meanings to the same learning initiative.
- Retrospective orientation. Interpretation draws on previous actions and experiences. Reviewing earlier decisions and outcomes allows organizational members to reconsider established assumptions and incorporate experience into subsequent learning.
- Environmental enactment. Organizational members do not merely respond to their surroundings; their actions also influence the conditions they subsequently interpret. Learning practices and the environments in which they operate therefore develop through continuing interaction.
- Social embeddedness. Sensemaking takes place through communication and interaction with others. Dialogue, cooperation, and the exchange of interpretations contribute to the development of shared understanding and collective learning.
- Continuous process. Interpretation is not confined to isolated decisions or events. Organizational members continually revise their understanding as new information becomes available and circumstances change.
- Cue-based interpretation. Individuals attend selectively to information that they regard as relevant. Differences in responsibilities, experience, and organizational context may therefore lead units to interpret the same information in different ways.
- Plausibility-driven interpretation. Sensemaking is concerned with developing interpretations that are sufficiently coherent to guide action, rather than establishing complete certainty. Such interpretations allow organizational members to proceed when information remains incomplete or ambiguous.

These properties suggest that learning capability cannot be understood solely through the presence or absence of particular organizational practices. Their implementation also depends on how employees interpret those practices, communicate their implications, and relate them to existing knowledge. For example, an initiative intended to encourage experimentation may have different consequences in a unit where employees openly discuss unsuccessful attempts and one where such discussions are discouraged. Similarly, knowledge transfer procedures may be less effective when employees do not share an understanding of how the transferred knowledge relates to their responsibilities.

Khrupa [32] identified six sensemaking schemas in knowledge-intensive firms: customer-centricity, business model interpretation, confirmation-seeking behavior, higher-purpose orientation, framing through familiar objects, and stakeholder management. These schemas illustrate how organizational members may interpret information according to the concerns and activities of their working environments. Units dealing with customers, internal operations, or strategic planning may therefore approach the same learning initiative from different perspectives. This variation provides a conceptual reason for allowing the expected contribution of an OLC dimension to differ across organizational units.

Weick [14] also examined sensemaking across several levels, including individual interpretation, interpersonal construction of shared meaning, formal organizational structures, and broader cultural or institutional contexts. This multilevel perspective is particularly relevant to the allocation of learning capabilities. Although individuals interpret information, many learning activities depend on communication, coordination, and shared practices at the unit level. The present study therefore represents sensemaking through unit-specific conditions associated with dialogue, shared mental models, collaborative culture, and organizational identification. These conditions provide the conceptual basis for the sensemaking effectiveness coefficient introduced in Section 3. A separate minimum capacity requirement is used to represent the need for sufficient collective interpretation and communication within each unit. These mathematical representations are operational assumptions derived from the theoretical discussion; their empirical validity requires further examination.

2.3 Organizational Learning Capability Dimensions

The allocation model considers ten OLC dimensions drawn from established measurement approaches. Together, they address organizational conditions associated with knowledge acquisition, interpretation, exchange, integration, and retention. The dimensions are discussed below in relation to their relevance to organizational learning and the decisions involved in allocating learning activities across units.

Risk Taking. Risk taking concerns the willingness to act under uncertainty, consider unfamiliar approaches, and accept the possibility of unsuccessful outcomes. Hedberg [33] associated organizational learning with the capacity to question established practices and respond to changing conditions. Sitkin [34] examined how organizations may learn from relatively small failures when these experiences are recognized and examined rather than disregarded. In this context, risk taking is not simply a preference for uncertain decisions; it also concerns the conditions under

which employees can discuss errors and reconsider previous assumptions. Research on organizational learning, psychological safety, and technological change has continued to examine these relationships [28, 35]. From a sensemaking perspective, the relevance of risk taking may depend on how a unit interprets mistakes and whether employees regard experimentation as an accepted part of their work.

Interaction with the External Environment. Interaction with the external environment allows organizations to obtain information, ideas, and experience from customers, suppliers, competitors, research institutions, and other external actors. Hedberg [33] emphasized the importance of environmental conditions in prompting organizations to reconsider existing knowledge and practices. External interaction is particularly relevant when established routines no longer provide adequate responses to changing circumstances. However, information obtained from outside the organization must still be interpreted and assessed before it can inform internal decisions. Units with frequent external contact may therefore require different learning priorities from units whose activities are primarily internal. Digital communication channels and interorganizational relationships provide additional means of obtaining and exchanging such knowledge.

Dialogue. Dialogue provides opportunities for organizational members to exchange information, examine assumptions, and develop common interpretations of their experiences. Schein [36] regarded dialogue as an important means of developing mutual understanding within organizations. Dibella et al. [25] also emphasized the role of everyday interaction in organizational learning. Through discussion, employees can compare individual observations, identify differences in interpretation, and relate new information to established working practices. Dialogue is consequently relevant to both knowledge sharing and collective sensemaking. Its organizational significance extends to remote and hybrid working arrangements, where communication increasingly takes place through digital systems [4, 12]. Within the proposed allocation model, dialogue is one of the dimensions associated with the collective interpretation of knowledge.

Participative Decision Making. Participative decision-making concerns the involvement of employees in decisions affecting their work and the wider organization. Participation allows knowledge held by individuals at different organizational levels to be considered during decision processes. Scott-Ladd and Chan [37] examined the relationship between employee participation, decision-making, and organizational learning. When employees contribute to decisions, they may also develop a clearer understanding of the reasons for organizational actions and the information on which those actions depend. The growing use of digital technologies and analytical tools has introduced further possibilities for incorporating distributed knowledge into managerial decisions [19]. From a sensemaking perspective, participation provides a setting in which differing interpretations can be discussed before decisions are implemented.

Managerial Commitment. Managerial commitment concerns the extent to which organizational leaders recognize learning as an organizational priority and provide the direction and resources needed to support it. Management influences learning through resource allocation, communication, and the establishment of expectations regarding knowledge acquisition and exchange. The importance of leadership and shared organizational direction has been discussed in established studies of learning capability and organizational development [23, 38]. Managers also participate in sensegiving by communicating interpretations of organizational change and influencing how employees understand new initiatives [39]. Research on digital transformation has further examined the role of management in developing organizational capabilities under changing technological conditions [8]. Managerial commitment is therefore relevant both to the availability of learning resources and to the interpretation of learning objectives within organizational units.

Systems Perspective. A systems perspective concerns the ability of organizational members to understand their activities as part of a wider organizational structure. Employees and departments need to recognize how their responsibilities relate to shared objectives and how decisions in one area may affect other areas [23]. This perspective is closely associated with the development of shared mental models, through which organizational members establish common interpretations of organizational processes and interdependencies. In knowledge management, a systems perspective helps connect knowledge generated within individual units to the requirements of the organization as a whole. It is particularly relevant when learning activities depend on coordination across functional boundaries.

Openness and Experimentation. Openness and experimentation concern the willingness to consider unfamiliar ideas, question established procedures, and test alternative approaches to organizational problems. These practices allow organizations to examine knowledge obtained from both internal experience and external sources. Their contribution to learning depends partly on whether employees can discuss proposed changes, assess experimental outcomes, and incorporate the resulting knowledge into subsequent activities. Recent studies of organizational learning and technological change have considered experimentation in relation to digital transformation, organizational adaptation, and changing management practices [4, 28]. From a sensemaking perspective, openness provides opportunities for employees to reconsider existing interpretations, while experimentation generates experience that can inform further learning.

Knowledge Transfer and Integration. Knowledge transfer and integration concern the movement of knowledge

among organizational members and its incorporation into collective practices. Transfer may occur through discussion, documentation, training, and cooperation across units, while integration involves relating newly acquired knowledge to existing procedures, routines, and organizational memory. These processes depend on the ability of employees to understand and use knowledge developed by others. Established research on organizational learning has emphasized the importance of organizational arrangements that support knowledge sharing and retention [23]. From a sensemaking perspective, transferred information becomes useful when recipients can interpret its relevance to their own activities. This dimension is therefore directly connected to the knowledge management purpose of the proposed allocation model.

Teamwork and Group Problem Solving. Teamwork concerns the coordination of employees who combine their knowledge and experience to address shared tasks or problems. Group problem solving provides opportunities for members to examine alternative interpretations, identify relevant information, and develop responses that may not emerge from individual work alone. These activities can support knowledge transfer across functional and professional boundaries. The increasing use of remote and hybrid working arrangements has also drawn attention to the conditions under which geographically dispersed employees exchange knowledge and coordinate their activities [12]. Research on organizational learning has examined the importance of collective practices in these settings [4]. In the present study, teamwork and group problem solving are particularly relevant to the representation of collective sensemaking capacity.

Demonstration of Mission and Goals. This dimension concerns the extent to which organizational members understand the organization's purpose, objectives, and expectations regarding their own contributions. Clearly communicated goals provide a common reference point for interpreting decisions and evaluating learning activities. Senge [38] discussed the importance of shared vision in organizational learning, particularly in connecting individual efforts with broader organizational aspirations. When employees understand how their work relates to organizational objectives, they may be better positioned to interpret new information and coordinate their actions with those of other units. The demonstration of mission and goals is therefore relevant to organizational identification and the development of shared understanding.

Table 1 presents the measurement items associated with these ten dimensions and identifies their literature sources. The items provide a basis for describing the organizational practices represented by each dimension. In the mathematical model, the dimensions are treated as candidate areas for implementation, while their expected contributions and costs are specified separately for individual organizational units. This distinction allows the allocation problem to account for differences in learning requirements and sensemaking conditions without assuming that every dimension has the same importance throughout the organization.

3 Problem Definition and Mathematical Modelling with Sensemaking Integration

OLC comprises several dimensions whose contributions to organizational learning may differ across units. These differences arise from variations in operational responsibilities, learning requirements, organizational strategies, environmental conditions, and accumulated experience. Consequently, the implementation of learning capability dimensions cannot be determined solely by their general importance to an organization. Knowledge management teams must also consider where particular learning activities are likely to make the greatest contribution and whether the resources required for their implementation are available. Previous research has examined the relationship between organizational learning capabilities, resource allocation, and organizational performance [5, 18]. These considerations provide the basis for treating the selection of OLC implementation items as a constrained allocation problem.

The allocation problem involves determining which learning capability items should be implemented in each organizational unit while accounting for differences in their expected contributions and implementation costs. A uniform allocation may be inappropriate because organizational units do not necessarily have identical learning priorities or resource requirements. Moreover, the selection of learning activities must satisfy both unit-level and organization-wide constraints. Analytical approaches to knowledge management provide a means of examining such decisions systematically, particularly when several competing allocation alternatives must be considered [5, 18]. The use of quantitative decision models is also consistent with research connecting data analytics, knowledge sharing, and managerial decision-making [19]. In the present study, the allocation problem is formulated as a binary optimization model in which implementation items are selected to maximize their combined weighted contribution to organizational learning.

Sensemaking introduces an additional consideration into this decision. The contribution of a learning activity may depend on how employees interpret its purpose, communicate relevant knowledge, and establish a shared understanding of the actions required. Khrupa [32] identified different sensemaking schemas in knowledge-intensive organizations, while Cristofaro [29] examined how sensemaking relates to organizational interpretation and adaptation. These perspectives suggest that the same learning capability dimension may have different implications across organizational units. To represent this variation, the proposed model incorporates two sensemaking-related

components. The first is an effectiveness coefficient that adjusts the contribution of each implementation item according to the sensemaking conditions associated with its organizational unit and OLC dimension. The second is a minimum capacity constraint that requires each unit to maintain a specified level of collective interpretation capability. Together, these components connect the theoretical discussion in Section 2 with the mathematical allocation problem. They are treated as model assumptions rather than empirically established relationships.

Table 1. Organizational learning capability dimensions and associated measurement items

Dimension	Measurement Items	Literature Source
Risk taking	People are encouraged to take risks in this organization	[26]
	People here often venture into unknown territory	[7]
Interaction with the external environment	It is part of the work of all staff to collect, bring back, and report information about what is going on outside the company	[27]
	There are systems and procedures for receiving, collating and sharing information from outside the company	[27]
	People are encouraged to interact with the environment: competitors, customers, technological institutes, universities, suppliers etc.	[27]
Dialogue	Employees are encouraged to communicate	[26]
	There is a free and open communication within my work group	[26]
	Managers facilitate communication	[27]
	Cross-functional teamwork is a common practice here	[23]
Participative decision making	Managers in this organization frequently involve employees in important decisions	[24]
	Policies are significantly influenced by the view of employees	[27]
	People feel involved in main company decisions	[27]
Team work and group problem solving	Current approach of the organization encourages personnel to solve the problems cooperatively, before discussing them with managers	[24]
	We often cannot form unofficial groups to solve the problems of the organization	[24]
	Majority of problem-solving groups are members of different operational environments	[24]
Demonstration of mission and goals	There is widespread support and acceptance of the Organization's mission statement	[24]
	I do not understand how the mission of the organization is to be achieved (r)	[24]
	The organization's mission statement identifies values to which all employees must conform	[24]
	We have opportunities for self-assessment with respect to goal attainment	[24]
Managerial commitment	The managers frequently involve their staff in important decision-making processes	[2]
	Employee learning is considered more of an expense than an investment	[2]
	The firm's management looks favorably on carrying out changes in any area to adapt to and/or keep ahead of new environmental situations	[2]
	Employee learning capability is considered a key factor in this firm	[2]
	In this firm, innovative ideas that work are rewarded	[2]
Systems perspective	All employees have generalized knowledge regarding this firm's objectives	[2]
	All parts that make up this firm (departments, sections, work teams, and individuals) are well aware of how they contribute to achieving the overall objectives	[2]
	All parts that make up this firm are interconnected, working together in a coordinated fashion	[2]
Openness and experimentation	This firm promotes experimentation and innovation as a way of improving the work processes	[2]
	This firm follows up what other firms in the sector are doing, adopting those practices and techniques it believes to be useful and interesting	[2]
	Experiences and ideas provided by external sources (advisors, customers, training firms, etc.) are considered a useful instrument for this firm's learning	[2]
	Part of this firm's culture is that employees can express their opinions and make suggestions regarding the procedures and methods in place for carrying out tasks	[2]
Knowledge transfer and integration	Errors and failures are always discussed and analyzed in this firm, on all levels	[2]
	Employees have the chance to talk among themselves about new ideas, programs, and activities that might be of use to the firm	[2]
	In this firm, teamwork is not the usual way to work	[2]
	The firm has instruments (manuals, databases, files, organizational routines, etc.) that allow what has been learnt in past situations to remain valid, although the employees are no longer the same	[2]

3.1 Mathematical Formulation

The proposed model considers a set of organizational units, OLC dimensions, and implementation items associated with each dimension. The allocation decision concerns whether a particular implementation item should be selected for a given organizational unit.

Let I denote the set of OLC dimensions, J the set of organizational units, and K_i the set of candidate implementation items associated with dimension i. The binary decision variable x_{ijk} indicates whether implementation

item k of dimension i is selected in unit j.

The objective is to maximize the total sensemaking-adjusted contribution of the selected implementation items across all organizational units:

$$\max \sum_{j} \sum_{i} \sum_{k} w^{*}_{ijk} x_{ijk} \tag{1}$$

where, w^{*}_{ijk} represents the sensemaking-adjusted weight of implementation item k belonging to dimension i in organizational unit j. The decision variable x_{ijk} equals 1 when the corresponding item is selected and 0 otherwise.

The objective function evaluates the combined contribution of the selected items under the specified model parameters. It does not directly measure observed improvements in organizational performance. Rather, the resulting objective value represents the estimated learning capability contribution associated with a feasible allocation.

The model is subject to minimum sensemaking capacity, implementation cost, budget, and minimum learning capability requirements, as specified in Sections 3.4–3.7.

3.2 Sensemaking-Adjusted Weight Calculation

The contribution of an OLC implementation item is assumed to depend partly on the sensemaking conditions under which it is introduced. Organizational units may differ in their communication practices, shared mental models, collaborative relationships, and identification with organizational objectives. These differences may influence how learning activities are interpreted and incorporated into established practices.

To represent this relationship, a sensemaking effectiveness coefficient s_{ij} is introduced for dimension i in organizational unit j. The base contribution of an implementation item is adjusted as follows:

$$w^{*}_{ijk} = w_{ijk} \left(1 + s_{ij}\right) \tag{2}$$

where, w_{ijk} is the base weight of implementation item k of dimension i in unit j, and s_{ij} is the corresponding sensemaking effectiveness coefficient, with $0 \leq s_{ij} \leq 1$.

Under this specification, the sensemaking coefficient acts as a nonnegative multiplier of the base contribution. A value of zero leaves the base weight unchanged, whereas larger values increase its contribution to the objective function. This functional form is an assumption of the proposed model. It does not imply that an equivalent proportional increase in organizational learning outcomes has been empirically observed.

Four factors are considered in estimating s_{ij}:

- D_{ij}: Dialogue and open communication associated with dimension i in unit j.
- M_{ij}: Shared mental models relevant to the implementation of dimension i in unit j.
- C_{ij}: Collaborative culture and social interaction supporting the implementation of dimension i in unit j.
- I_{ij}: Organizational identification and alignment with the objectives associated with dimension i in unit j.

These factors reflect the theoretical discussion of sensemaking in Section 2.2. Their inclusion provides an operational representation of selected sensemaking conditions rather than a comprehensive measurement of the sensemaking construct.

A structured scoring procedure is specified to estimate the coefficient. Each factor is assessed on a five-point Likert scale, ranging from 1 (very low) to 5 (very high). The raw rating $r_{f,ij}$, where $f \in \{D, M, C, I\}$, is normalized to the interval $[0, 1]$:

$$\hat{n}_{f,ij} = \frac{r_{f,ij} - 1}{4} \tag{3a}$$

When several assessors participate, their normalized ratings are averaged. For example, assessments may be provided by the unit manager, a knowledge management representative, and a senior manager familiar with the unit's activities. The mean normalized score is calculated as:

$$\bar{n}_{f,ij} = \frac{1}{n} \sum_{p=1}^{n} \hat{n}^{(p)}_{f,ij} \tag{3b}$$

where, n is the number of assessors and $\hat{n}^{(p)}_{f,ij}$ is the normalized rating assigned by assessor p to factor f for dimension i in unit j.

Averaging the assessments provides a common numerical estimate when multiple managerial perspectives are available. However, it does not eliminate differences in interpretation or guarantee measurement reliability. Where the procedure is applied to observed organizational data, agreement among assessors should be examined.

The four factor scores are combined using a weighted additive formulation:

$$s_{ij} = w_D \bar{n}_{D,ij} + w_M \bar{n}_{M,ij} + w_C \bar{n}_{C,ij} + w_I \bar{n}_{I,ij} \tag{3c}$$

where, w_D, w_M, w_C, and w_I represent the relative importance assigned to the four sensemaking factors, subject to:

$$\sum_{f \in \{D,M,C,I\}} w_f = 1, \quad w_f \geq 0. \tag{3d}$$

When no factor is assigned greater importance, equal weights of $w_f = 0.25$ may be used.

The resulting coefficient represents an assessed level of sensemaking support for the implementation of a particular OLC dimension within an organizational unit. Its numerical value depends on the ratings and factor weights selected by the assessors. The scoring procedure makes the calculation explicit and repeatable, although its measurement validity and sensitivity to alternative weighting schemes require further empirical examination.

3.3 Base Weight Calculation

The base weight w_{ijk} represents the estimated contribution of implementation item k of OLC dimension i to learning capability in organizational unit j, before the sensemaking adjustment is applied. Its calculation draws on the measurement items associated with the OLC dimensions presented in Table 1.

A distinction is made between measurement items and implementation items. Measurement items describe organizational practices or conditions associated with learning capability, whereas implementation items represent activities or measures that may be selected within an organizational unit. The relationship between these two sets of items is used to estimate the contribution of each candidate implementation item.

Let k' denote a measurement item associated with the relevant OLC dimension. The parameter $N_{ijkk'}$ represents the assessed relationship between measurement item k' and implementation item k of dimension i in unit j. Its value ranges from 0 to 100 and is specified by the organization's knowledge management team.

A higher value of $N_{ijkk'}$ indicates a stronger assessed relationship between the measurement item and the corresponding implementation item. These values provide inputs to the base-weight calculation; they should not be interpreted as observed improvements in organizational learning unless supported by empirical measurements.

The contribution associated with each measurement item is represented by $w_{ijkk'}$. This quantity is calculated using a parameterized function involving the coefficients a, b, c, d, α, β, and γ, whose values are determined by the knowledge management team:

$$w_{ijkk'} = \begin{cases} 0, & N_{ijkk'} < a, \\ \alpha, & a \leq N_{ijkk'} < b, \\ \beta, & b \leq N_{ijkk'} < c, \\ \gamma, & c \leq N_{ijkk'} < d, \\ 100, & N_{ijkk'} \geq d. \end{cases} \tag{4}$$

The thresholds a, b, c, and d divide the assessed relationship scores into five intervals, while α, β, and γ specify the contribution weights assigned to the three intermediate intervals. The highest interval is assigned a weight of 100. The base weight of implementation item k in dimension i and unit j is then obtained by aggregating the contributions of its associated measurement items:

$$w_{ijk} = \sum_{k'} w_{ijkk'}. \tag{5}$$

This formulation allows the contribution of an implementation item to reflect several measurement items associated with the same OLC dimension.

The estimated base weights are subsequently adjusted using Eq. (2). Thus, the model distinguishes between the contribution assigned to an implementation item through the OLC measurement structure and the additional effect assumed to arise from the sensemaking conditions of the organizational unit.

For practical implementation, the measurement-to-implementation relationships and the parameters used in Eq. (4) must be specified consistently across organizational units. Their values should also be documented to permit independent reproduction of the optimization results.

3.4 Sensemaking Capability Constraint

The sensemaking effectiveness coefficient adjusts the expected contribution of selected learning activities, but it does not independently ensure that an organizational unit possesses sufficient capacity for collective interpretation and communication. A separate constraint is therefore introduced to represent the minimum sensemaking capability required within each unit.

The constraint is expressed as:

$$\sum_i \sum_k t_{ijk} \cdot x_{ijk} \geq S_j^{\min}, \quad \forall j \tag{6}$$

where, t_{ijk} represents the estimated contribution of implementation item k of dimension i to sensemaking capability in unit j, and $S_j^{\min}$ is the minimum required sensemaking capability for that unit.

The parameter t_{ijk} is associated with measurement items relevant to collective interpretation and communication. These include items concerning dialogue, employee participation in decision-making, teamwork and group problem solving, and openness to alternative ideas and practices.

The constraint requires the combined sensemaking contribution of selected implementation items to meet or exceed the specified threshold in each organizational unit. It therefore prevents the model from selecting an allocation whose estimated sensemaking capacity falls below the required level.

The theoretical basis for this requirement lies in the role of communication and shared interpretation in organizational learning. Sensemaking research emphasizes that organizational members must interpret information and develop workable understandings before knowledge can be incorporated into collective action [14, 16].

The additive formulation assumes that the contributions of individual implementation items can be aggregated to represent unit-level sensemaking capacity. The values of t_{ijk} and $S_j^{\min}$ are model parameters and require an explicit assessment procedure when the framework is applied in practice. The constraint should therefore be understood as a formal representation of a minimum organizational requirement rather than evidence that the selected activities will necessarily produce the corresponding level of sensemaking capability.

3.5 Budget and Cost Constraints

The implementation of OLC activities requires financial and organizational resources. Because these resources are limited, the selection of learning capability items must account for their expected contributions and implementation costs.

Let C_{ijk} denote the cost of implementing item k of dimension i in organizational unit j. Each unit is assigned a maximum implementation budget A_j. The corresponding constraint is:

$$\sum_i \sum_k C_{ijk} x_{ijk} \leq A_j, \quad \forall j. \tag{7}$$

This constraint ensures that the combined implementation cost of the selected learning capability items does not exceed the budget available to each organizational unit.

In addition to unit-level restrictions, the total expenditure across the organization must remain within the overall budget B:

$$\sum_{i \in I} \sum_{j \in J} \sum_{k \in K_i} C_{ijk} x_{ijk} \leq B. \tag{8}$$

The two constraints represent different levels of financial control. Eq. (7) restricts expenditure within individual units, whereas Eq. (8) limits the total expenditure across all units.

Both constraints are necessary when organizational units have separately assigned budgets and the organization also imposes an overall spending limit. Their inclusion allows the model to identify allocations that satisfy local financial restrictions while remaining feasible at the organizational level.

The resulting allocation depends on the estimated costs and available budgets. These parameters must therefore be defined using a consistent cost basis when the model is applied to organizational data.

3.6 Minimum Effect Constraint

Maximizing the total weighted contribution of selected OLC items does not necessarily ensure that every organizational unit receives an adequate allocation. Under limited resources, an unconstrained objective may favor units in which implementation items have comparatively high estimated contributions.

To represent minimum learning capability requirements at the unit level, the following constraint is imposed:

$$\sum_i \sum_k w_{ijk}^* x_{ijk} \geq M_j, \quad \forall j \tag{9}$$

where, M_j denotes the minimum required sensemaking-adjusted learning capability contribution in organizational unit j.

The constraint requires the selected implementation items in each unit to achieve a combined weighted contribution of at least M_j. It therefore establishes a minimum allocation requirement independently of the objective function's organization-wide maximization criterion.

The threshold may affect the number and combination of implementation items selected in a unit. A higher requirement can restrict the feasible allocation alternatives and may require additional expenditure, depending on the contribution and cost of the available items. However, increasing the threshold does not necessarily lead to a

proportional increase in expenditure, since the resulting allocation also depends on the available combinations of implementation items.

Eqs. (7)–(9) jointly represent the relationship between learning capability requirements and financial feasibility. An allocation is feasible only when the minimum weighted contribution can be achieved without exceeding the relevant unit-level and organization-wide budgets.

The minimum effect threshold is a planning parameter rather than a directly observed measure of organizational learning performance. Its selection should therefore reflect the organization's stated learning requirements and the scale of the estimated contribution weights.

3.7 Binary Constraint

The selection of OLC implementation items is represented by binary decision variables, defined as follows:

$$x_{ijk} \in \{0, 1\}, \quad \forall i, j, k \tag{10}$$

where, $x_{ijk} = 1$ indicates that implementation item k associated with OLC dimension i is selected for organizational unit j, whereas $x_{ijk} = 0$ indicates that the item is not selected.

The binary formulation reflects the discrete nature of the allocation decisions considered in this study. Each implementation item is either included in or excluded from the learning capability development plan of a particular organizational unit. Together with the objective function and the constraints specified in Eqs. (1)–(9), this formulation defines a binary linear optimization model, provided that all coefficients are determined before optimization.

The model identifies feasible combinations of OLC implementation items across organizational units while accounting for their sensemaking-adjusted contributions, implementation costs, and minimum capability requirements. It provides knowledge management teams with a consistent basis for comparing allocation alternatives under different organizational conditions. The binary formulation assumes that implementation items are indivisible within the planning period; partial implementation and variations in implementation intensity are not explicitly represented.

4 Computational Results

The proposed mathematical decision-support model was examined using a hypothetical organization consisting of five units. Ten OLC dimensions were considered, with three measurement items and three implementation items specified for each dimension. The computational experiment was designed to examine how learning capability items were allocated under different organizational requirements, budget restrictions, and sensemaking conditions. As the analysis relied on assumed parameter values rather than observations from an actual organization, its purpose was to examine the behavior of the optimization model rather than to establish its effectiveness in organizational practice.

The coefficients required for calculating the base contribution weights were specified as part of the experimental setup. These included α, β, γ, a, b, c, and d in Eq. (4), together with the implementation costs and the parameters used to assess the contributions of individual learning capability items. The experimental setting specified relationship scores within the range of 10–87. The resulting binary optimization problem was solved using LINGO (a commercial optimization software package), and the selected implementation items were recorded for each organizational unit.

Three sets of parameters were used to represent sensemaking conditions. The sensemaking effectiveness coefficients s_{ij} were assigned values between 0.1 and 0.8 to reflect differences across organizational units and learning capability dimensions. The minimum sensemaking capacity requirement $S_j^{\min}$ was set at 50 for each unit, while the sensemaking contribution coefficients t_{ijk} ranged from 5 to 25. These values were treated as assumed inputs to the computational experiment rather than empirically estimated measures of organizational sensemaking.

Table 2 presents the unit-level budget limits, minimum learning capability requirements, and minimum sensemaking capacity thresholds. The organization-wide budget was set at USD 70,000. Although the sum of the unit-level budget limits exceeded this amount, the overall budget constraint restricted the combined expenditure across all five units. The two budget constraints therefore represented different levels of financial control within the allocation problem.

Table 2. Budget limits and minimum capability requirements for the five organizational units

Unit j	Unit Budget A_j (USD)	Minimum Learning Effect M_j	Minimum Sensemaking Capacity $S_j^{\min}$
1	16,000	70	50
2	14,000	75	50
3	24,000	70	50
4	14,000	70	50
5	18,000	80	50

Note: j = organizational unit index. A_j is expressed in USD. M_j and $S_j^{\min}$ are dimensionless effect and capacity scores, respectively, as defined in Sections 3.4 and 3.6. The organization-wide budget constraint is $B =$ USD 70,000.

The coefficients used in the base-weight calculation are reported in Table 3. These parameters were specified for the computational experiment and held constant when examining changes in the sensemaking and budget parameters. Their values represent the numerical test setting and should not be interpreted as empirical estimates of organizational behavior.

Table 3. Coefficients used in the base-weight calculation

Coefficient	α	β	γ	a	b	c	d
Value	27	45	67	13	38	53	79

Note: a, b, c, and d denote the threshold values for partitioning the assessed relationship scores into intervals, while α, β, and γ represent the contribution weights assigned to the three intermediate intervals in Eq. (4).

The allocation results obtained from LINGO are summarized in Table 4. The table presents the selected OLC dimensions (i) for each organizational unit (j). A value of 1 indicates that at least one implementation item associated with dimension i was selected for unit j, and a value of 0 indicates that no items from that dimension were selected. The table records the reported allocation under the base-case parameter setting.

Table 4. Allocation of organizational learning capability dimensions across organizational units ($i \times j$ matrix)

Organizational Learning Capability Dimension (i)	**Unit 1 ($j=1$)**	**Unit 2 ($j=2$)**	**Unit 3 ($j=3$)**	**Unit 4 ($j=4$)**	**Unit 5 ($j=5$)**
1. Risk taking	0	1	1	0	0
2. Interaction with the external environment	1	0	1	0	1
3. Dialogue	1	1	0	1	0
4. Participative decision making	1	0	0	1	1
5. Managerial commitment	0	1	0	1	0
6. Systems perspective	1	1	0	0	0
7. Openness and experimentation	1	0	1	0	0
8. Knowledge transfer and integration	0	0	0	1	1
9. Teamwork and group problem solving	1	1	0	0	0
10. Demonstration of mission and goals	0	0	1	0	1
Total selected	6	5	4	4	4

Note: i = organizational learning capability dimension index; j = organizational unit index. A value of 1 indicates that the dimension is selected for the corresponding unit, and 0 indicates that it is not selected.

The selected implementation items differed across the five organizational units. Unit 1 contained selected implementation items from dimensions 2, 3, 4, 6, 7, and 9, whereas Unit 2 contained items from dimensions 1, 3, 5, 6, and 9. Unit 3 included items from dimensions 1, 2, 7, and 10. The allocation for Unit 4 comprised dimensions 3, 4, 5, and 8, while Unit 5 contained selected items from dimensions 2, 4, 8, and 10.

These differences indicate that the reported solution did not assign identical combinations of learning capability items to every unit. Instead, the selected configurations varied according to the contribution coefficients, implementation costs, and constraints specified in the computational setting. The results are consistent with the model's purpose of representing differences in organizational learning requirements and sensemaking conditions within a common allocation framework.

The allocation patterns also provide a basis for examining how learning capability priorities may differ among organizational units. Rather than treating all dimensions as equally relevant to every unit, the model permits different combinations of implementation items to be selected under the specified resource limits. In knowledge management terms, this approach allows learning-related investments to be considered alongside differences in organizational conditions, although the resulting allocations do not establish that knowledge sharing or organizational performance would improve in practice.

The computational results illustrate the allocation decisions reported for the base-case setting. However, the summarized allocation does not provide the objective function value, actual expenditure within each unit, or the remaining margins above the minimum learning and sensemaking requirements. These quantities must be obtained from the corresponding numerical inputs and solver output before the feasibility and optimality of the reported solution can be independently verified.

4.1 Sensitivity Analysis

A sensitivity analysis was conducted to examine how changes in selected model parameters affected the allocation of OLC implementation items. Three parameters were considered: the sensemaking effectiveness coefficient s_{ij}, the minimum sensemaking capacity requirement $S_j^{\min}$, and the organization-wide budget B. Each parameter was varied separately by -20%, -10%, +10%, and +20% relative to its base-case value. The reported allocation changes were then compared with the base-case configuration presented in Table 4.

Sensemaking effectiveness coefficient. Changes in s_{ij} affected the contribution of implementation items through the sensemaking-adjusted weights defined in Eq. (2). According to the reported sensitivity results, increases of 10% and 20% left the allocation patterns for Units 1, 3, 4, and 5 unchanged. At the 20% increase, the allocation for Unit 2 changed, with an implementation item associated with dimension 4 replacing an item associated with dimension 1. This result suggested that changes in sensemaking conditions could alter the relative priority assigned to certain learning capability items.

Reductions in s_{ij} also produced changes in the reported allocation. Units 1 and 4 retained their original configurations, while Unit 5 lost a selected implementation item associated with dimension 10 under the reduced-coefficient scenarios. The allocation for Unit 2 was reported to return to its base-case configuration. These observations suggested that the influence of sensemaking adjustments was not uniform across organizational units. Instead, the response depended on the relative contribution weights of the available implementation items and the constraints governing their selection.

Minimum sensemaking capacity requirement. The second analysis examined changes in $S_j^{\min}$. Increasing the minimum requirement by 10% and 20% reportedly left the allocations for Units 1, 3, and 5 unchanged. Unit 4 selected an additional implementation item associated with dimension 3 under the tighter capacity requirement. This change was consistent with the purpose of the minimum sensemaking constraint, which restricts solutions that provide insufficient contributions to collective interpretation and communication.

Reducing $S_j^{\min}$ by 10% and 20% reportedly produced no changes in the selected implementation items. Within the examined scenarios, the base-case allocation therefore appeared to remain preferred when the minimum sensemaking requirement was relaxed. However, the absence of an allocation change does not establish the numerical slack of the constraint. This must be determined from the corresponding constraint values in the optimization results.

Organization-wide budget. The third analysis considered changes in the total budget B. A 10% reduction reportedly removed an implementation item associated with dimension 8 from the allocation for Unit 4. A further reduction to 20% was reported to remove an item associated with dimension 9 from Unit 2. These changes suggested that the allocation of certain learning capability items was sensitive to tighter financial restrictions.

In contrast, increases of 10% and 20% in the total budget reportedly left the allocation unchanged. This observation may indicate that restrictions other than the organization-wide budget limited the selection of additional implementation items. Nevertheless, an unchanged allocation does not identify which constraints were binding. Confirmation requires the objective function values, actual budget utilization, and constraint slack obtained from the optimization solver.

Overall, the sensitivity results indicated that changes in sensemaking effectiveness coefficients and financial limits could alter the selected combinations of learning capability items. Relaxing the minimum sensemaking requirement did not change the reported allocation, whereas tightening this requirement affected the selection of implementation items in at least one organizational unit. The responses differed across units, indicating that the consequences of parameter changes depended on the interaction between learning requirements, contribution weights, and implementation costs.

From a knowledge management perspective, these findings illustrate the value of examining alternative parameter settings rather than relying on a single allocation. Differences in sensemaking conditions may influence which learning activities receive priority, while financial restrictions may require changes to the selected implementation items. The results also indicate the importance of carefully specifying the sensemaking-related coefficients, since these parameters directly affect the contribution assigned to learning capability items.

The analysis remains subject to the limitations of the computational setting. The reported changes were obtained using assumed parameter values, and the full numerical outputs for the sensitivity scenarios were not available for independent verification. Accordingly, the findings should be interpreted as preliminary evidence of the model's response to parameter variation rather than as confirmation of its effectiveness in actual organizational settings.

5 Discussion

5.1 Interpretation of Results Through a Sensemaking Lens

The computational results in Table 4 show that the reported allocation of OLC implementation items differed across the five organizational units. These differences arose within a common optimization framework that considered learning capability contributions, implementation costs, and sensemaking-related requirements. The results do not establish that the selected combinations would produce superior learning outcomes in practice. They do, however, provide a basis for examining how different learning priorities may be represented within a knowledge management decision-support model.

Unit 1. The reported solution selected implementation items associated with dimensions 2, 3, 4, 6, 7, and 9. This combination includes activities related to external knowledge acquisition, dialogue, participative decision-making, organizational coordination, and learning-oriented experimentation, depending on the dimension definitions adopted in Section 2.3. From a sensemaking perspective, these activities concern both the interpretation of information

and the communication of knowledge across organizational boundaries. The assumed sensemaking coefficient of $s_{ij} = 0.65$, reported for this unit in the original discussion, suggests that sensemaking conditions were intended to have a relatively strong influence on the adjusted contribution weights. This interpretation is consistent with the attention given to managerial sensemaking in organizational learning by Cristofaro [29] and to organizational capabilities under digital transformation by Warner and Wäger [8]. Nevertheless, the coefficient represents an assumed model input rather than an independently measured level of organizational maturity.

Unit 2. Implementation items associated with dimensions 1, 3, 5, 6, and 9 were selected. These dimensions cover activities connected with risk taking, dialogue, managerial involvement, organizational coordination, and knowledge-related practices. Their joint selection may be interpreted in relation to the social processes through which employees exchange information and establish common interpretations. Weick [14] identified interpersonal interaction as an important level of sensemaking, while Hermannrud and Haukåsen [30] examined the role of sensemaking in workplace learning. Participative practices and communication may also affect how organizational knowledge enters decision processes, as discussed by Scott-Ladd and Chan [37]. The reported sensitivity analysis indicated that Unit 2 responded to changes in the sensemaking effectiveness coefficients. This finding suggests that its allocation was sensitive to the relative contributions assigned to competing implementation items, although it does not establish that Unit 2 would function as a knowledge-sharing hub in an actual organization.

Unit 3. The reported allocation included implementation items associated with dimensions 1, 2, 7, and 10. These dimensions represent a combination of risk taking, interaction with the external environment, openness and experimentation, and the communication of organizational mission and goals. The configuration can be interpreted as one in which learning from unfamiliar information is considered alongside the need to relate new knowledge to broader organizational objectives. The assumed sensemaking coefficient of s_{ij} = 0.40, reported in the original discussion, indicates a lower specified level of sensemaking adjustment than in some other units. However, the numerical coefficient alone does not establish the actual maturity of the unit or the effectiveness of its learning practices. The selected combination instead illustrates how the model can assign different learning-related activities to a unit under its specified contribution weights and constraints.

Unit 4. The reported solution selected items associated with dimensions 3, 4, 5, and 8. These dimensions concern dialogue, participative decision-making, managerial commitment, and knowledge transfer and integration. Their combination is relevant to collective interpretation because communication and participation provide opportunities for employees to examine information from different perspectives. The minimum sensemaking capacity constraint is particularly relevant to this interpretation, although its actual contribution to the selected solution must be confirmed from the constraint values. Research by Edmondson and Bransby [35] on psychological safety and by Kniffin et al. [12] on changing workplace arrangements provides a broader context for understanding the organizational conditions under which knowledge exchange and collective learning take place. The reported sensitivity analysis also indicated that Unit 4 responded to tighter minimum sensemaking requirements, suggesting that this constraint may influence its allocation under alternative parameter settings.

Unit 5. The selected implementation items were associated with dimensions 2, 4, 8, and 10. This combination includes external interaction, participative decision-making, knowledge transfer and integration, and the demonstration of organizational mission and goals. From a knowledge management perspective, these activities concern the acquisition of information from outside the organization, its interpretation through collective decisions, and its incorporation into organizational practices. The assumed sensemaking coefficient of $s_{ij} = 0.55$ indicates the level of adjustment specified for this unit in the original computational discussion. The reported sensitivity analysis suggested that reductions in sensemaking effectiveness could alter the selection of items associated with organizational mission and goals. This observation is consistent with the model's representation of learning capability contributions as dependent partly on sensemaking conditions, although the organizational consequences of such changes remain untested.

The allocation patterns can be considered alongside previous research on the relationship between OLC and organizational performance. Migdadi [5] examined differences in the effects of learning capabilities on innovation and performance, while Goh et al. [18] addressed the relationship between learning capability and organizational outcomes. These studies support the relevance of examining learning capabilities in relation to organizational context, but they do not independently validate the allocation results reported here. Similarly, Ghasemaghaei [19] examined how knowledge sharing and analytical capabilities contribute to decision quality. The present study addresses a related but narrower problem by representing the selection of learning capability items as a constrained allocation decision. Its contribution lies in the formalization of this decision problem rather than in demonstrating changes in organizational performance.

5.2 Sensemaking as a Meta-Capability for Organizational Learning

Sensemaking may be regarded as a capability that influences how other organizational learning capabilities are interpreted and put into practice. Rather than constituting a separate learning activity alone, it concerns the processes

through which organizational members understand information, reconcile different interpretations, and coordinate their responses. In the proposed model, this relationship is represented through the sensemaking effectiveness coefficient and the minimum sensemaking capacity constraint. The formulation treats sensemaking as a condition affecting learning capability allocation, although the computational results do not establish its status as a meta-capability empirically.

First, sensemaking concerns the interpretation of learning initiatives within particular organizational settings. Gioia and Chittipeddi [39] examined how managers influence the meanings attached to organizational change through sensegiving. Whittle et al. [31] emphasized the role of language and discourse in organizational sensemaking, while Cristofaro [29] examined the relationship between sensemaking and organizational adaptation. Weick [14] also described sensemaking as a social process through which organizational members construct plausible interpretations of events. These perspectives provide a theoretical basis for allowing the contribution of a learning capability item to depend partly on the interpretive conditions of the unit in which it is implemented.

Second, sensemaking is closely associated with communication, dialogue, teamwork, and participative decision-making. Weick [14] emphasized the social character of sensemaking, while Hermannrud and Haukåsen [30] examined its role in workplace learning. Sousa and Rocha [4] considered learning processes in the context of organizational digital transformation, where knowledge is increasingly exchanged through technological systems. These studies suggest that the organizational conditions supporting shared interpretation may influence how learning practices are implemented. The proposed model represents this possibility by assigning sensemaking-related contribution coefficients to implementation items. The resulting allocation differences are consistent with the structure of the model, but they should not be interpreted as direct evidence that units with higher sensemaking coefficients achieved better learning outcomes.

Third, sensemaking provides a perspective on organizational responses to uncertainty. When established interpretations are no longer sufficient, employees must reconsider available information and develop explanations that allow coordinated action [14, 16]. Cristofaro [29] examined sensemaking in organizational adaptation, while Warner and Wäger [8] discussed the development of dynamic capabilities during digital transformation. In the present model, the minimum sensemaking capacity requirement ($S_j^{\min}$) represents an assumed threshold for collective interpretation and communication within each organizational unit. This constraint allows such requirements to be considered explicitly during resource allocation. It does not, however, demonstrate that the selected threshold is sufficient for effective sensemaking in an actual organization.

Fourth, sensemaking may contribute to the organizational processes associated with sensing changes, interpreting opportunities, and adjusting established practices. Research by Sousa and Rocha [4] on digital learning and Leonardi and Treem [40] on behavioral visibility in digitally mediated communication provides a broader context for understanding how information becomes available and meaningful within organizations. The proposed mathematical model addresses one aspect of this relationship by allowing sensemaking-related conditions to influence the selection of learning capability items. It therefore offers a formal representation of a theoretically motivated relationship between interpretation and resource allocation. Further empirical research is required to determine whether this representation corresponds to observed organizational learning processes.

5.3 Practical Implications

The proposed decision-support model has several implications for the planning and allocation of organizational learning resources. Its principal application is to help managers compare alternative combinations of learning capability items under specified financial and organizational constraints. The model does not prescribe a universally preferred configuration; its outputs depend on the learning requirements, contribution coefficients, implementation costs, and sensemaking conditions assigned to individual units.

For knowledge management teams, the model provides a structured means of examining which learning capability items should receive priority in different organizational units. The sensemaking effectiveness coefficients s_{ij} and contribution weights t_{ijk} require careful assessment because they influence the relative value assigned to implementation items and the feasibility of allocation decisions. Differences in communication practices, shared mental models, collaborative relationships, and organizational identification may be relevant when specifying these parameters. Goh and Richards [24] and Chiva et al. [27] emphasized the importance of systematic measurement in OLC research. Their measurement approaches provide a basis for identifying relevant organizational practices, although additional work is needed to establish how such measurements should be translated into the coefficients used in the optimization model.

For organizational leaders, the findings draw attention to the relationship between learning investments and the interpretation of organizational priorities. Managers influence how employees understand learning initiatives through communication, resource allocation, and the explanation of organizational objectives. This process is related to sensegiving, through which organizational members attempt to shape the interpretation of change [39]. Research by Whittle et al. [31] and Cristofaro [29] further illustrates the importance of interpretation in organizational processes.

In practical applications, managers should therefore consider whether the organizational conditions needed to support a selected learning activity are present before committing resources to its implementation.

For organizational design, the model allows differences among organizational units to be represented explicitly rather than assuming identical learning priorities throughout the organization. The multilevel character of sensemaking discussed by Weick [14] and the workplace learning processes examined by Hermannrud and Haukåsen [30] indicate that individual interpretation, group interaction, and organizational arrangements may affect how learning practices operate. The allocation model can assist in examining alternative distributions of learning activities across units, provided that its parameters reflect the organizational setting being considered.

For learning initiatives, the model suggests that resource allocation decisions should account for the communication and interpretation processes associated with implementation. Activities involving dialogue, participative decision-making, and teamwork may require different forms of organizational support from activities concerned primarily with acquiring external knowledge or maintaining organizational memory. Warner and Wäger [8] and Sousa and Rocha [4] discussed the development of organizational capabilities in changing technological environments. Their work provides a context for considering how learning investments may be coordinated with broader organizational development efforts.

For innovation-oriented learning, the allocation of experimentation, risk taking, knowledge transfer, and participative decision-making may be relevant to the conditions under which new ideas are generated and examined. The computational results included implementation items associated with these dimensions across several organizational units. However, the presence of such items in an allocation does not establish that the units would produce more innovations or achieve better innovation performance. The model instead provides a basis for examining how innovation-related learning activities might be distributed under different organizational priorities. Future applications should assess whether the selected configurations are associated with observable changes in knowledge use, experimentation, or innovation outcomes.

For sensemaking capability development, the minimum capacity constraint $S_j^{\min}$ provides a means of incorporating collective interpretation requirements into allocation decisions. Organizations considering this approach would need to establish appropriate capacity thresholds and determine how selected learning activities contribute to those thresholds. Research by Cristofaro [29] and Hermannrud and Haukåsen [30] provides theoretical support for examining sensemaking in organizational learning, but the numerical values used in the present model remain assumptions. Their practical use would require assessment and validation within the relevant organizational setting.

For strategic resource allocation, the model provides an analytical basis for comparing learning capability investments across organizational units. By changing budget limits, sensemaking coefficients, and minimum capability requirements, decision-makers can examine how alternative assumptions affect the selected implementation items. This form of scenario analysis may support discussions about learning priorities and resource distribution. Its value lies in making the assumptions and constraints underlying allocation decisions explicit, rather than in replacing managerial judgment or guaranteeing particular organizational outcomes.

6 Conclusions

This study developed a mathematical decision-support model for allocating OLC implementation items across organizational units under financial and learning-related constraints. The model addressed the allocation problem by considering the expected contribution of individual learning activities, differences in organizational requirements, and the resources available for implementation. Sensemaking theory was incorporated to represent how differences in collective interpretation and shared understanding may influence the contribution of learning capability items within particular organizational settings.

The study made two methodological contributions. First, the allocation problem was formulated as a binary optimization model in which learning capability items were selected according to their weighted contributions, implementation costs, unit-level budget limits, an organization-wide budget constraint, and minimum learning capability requirements. Second, sensemaking was incorporated through an effectiveness coefficient that adjusted the contribution of individual implementation items and a minimum capacity constraint representing requirements for collective interpretation and communication, as formalized in Eq. (6). This constraint requires that the selected implementation items in each unit provide at least the specified minimum estimated sensemaking contribution.

These components provide a formal means of considering sensemaking-related conditions within organizational learning resource allocation decisions.

The computational experiment considered a hypothetical organization comprising five units. The allocation results differed across units, indicating that the model selected different combinations of learning capability items under the specified parameters and constraints. Sensitivity analyses were also conducted to examine changes in sensemaking effectiveness coefficients, minimum sensemaking capacity requirements, and the organization-wide budget. The reported results indicated that changes in sensemaking coefficients and budget restrictions could alter selected implementation items, while relaxing the minimum sensemaking requirement did not change the allocation

in the examined scenarios. These findings describe the behavior of the model under assumed numerical conditions rather than observed changes in organizational learning performance.

From a theoretical perspective, the study connects OLC and sensemaking with a formal resource allocation problem. It represents sensemaking not only as a process relevant to the interpretation of knowledge but also as a condition that can be incorporated into the evaluation and selection of learning activities. From a practical perspective, the model offers knowledge management teams a structured basis for comparing unit-specific learning investments and examining alternative decisions under different resource limits and organizational requirements. Its application depends on the availability of suitable estimates for learning capability contributions, implementation costs, and sensemaking-related parameters.

Several limitations should be considered. The computational analysis relied on assumed parameter values and a limited hypothetical organizational setting. The allocation results have not been validated using observed organizational data, and the numerical outputs required for independent verification of all reported sensitivity scenarios were not available in the present analysis. In addition, the binary formulation treats implementation items as indivisible decisions and does not explicitly represent differences in implementation intensity or changes in learning capability over time. The model also assumes that the specified contribution coefficients adequately represent the relationships between selected learning activities and organizational learning requirements.

Future research should examine the model using data collected from actual organizations, including measurements of learning capability, sensemaking conditions, implementation costs, and knowledge management outcomes. Longitudinal applications would allow changes in learning requirements and sensemaking conditions to be considered over time. The model could also be extended to incorporate additional organizational characteristics, including leadership practices, organizational culture, industry conditions, and interdependencies among learning activities. Further work may examine how digital platforms, artificial intelligence, and analytical methods can support the measurement of relevant parameters and the evaluation of alternative allocation decisions.

The proposed approach provides a starting point for integrating sensemaking considerations into mathematical decisions concerning organizational learning resources. Its principal contribution is a structured representation of the allocation problem, while its practical validity and organizational consequences remain subjects for empirical investigation.

Author Contributions

Conceptualization, H.F. and M.A.; methodology, H.F.; software, F.A.; validation, M.A. and F.A.; formal analysis, H.F.; investigation, M.A.; resources, F.A.; data curation, H.F.; writing—original draft preparation, F.A.; writing—review and editing, H.F., M.A. and F.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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